

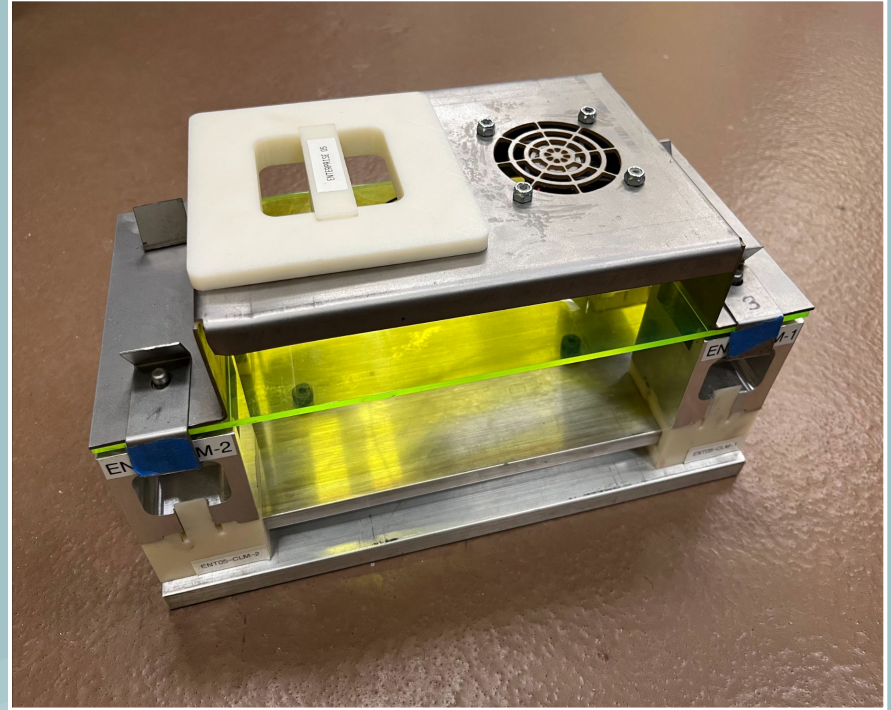
Enterprise 05 PM Closeout Presentation

By Ben Schaffer and Luke Morrison

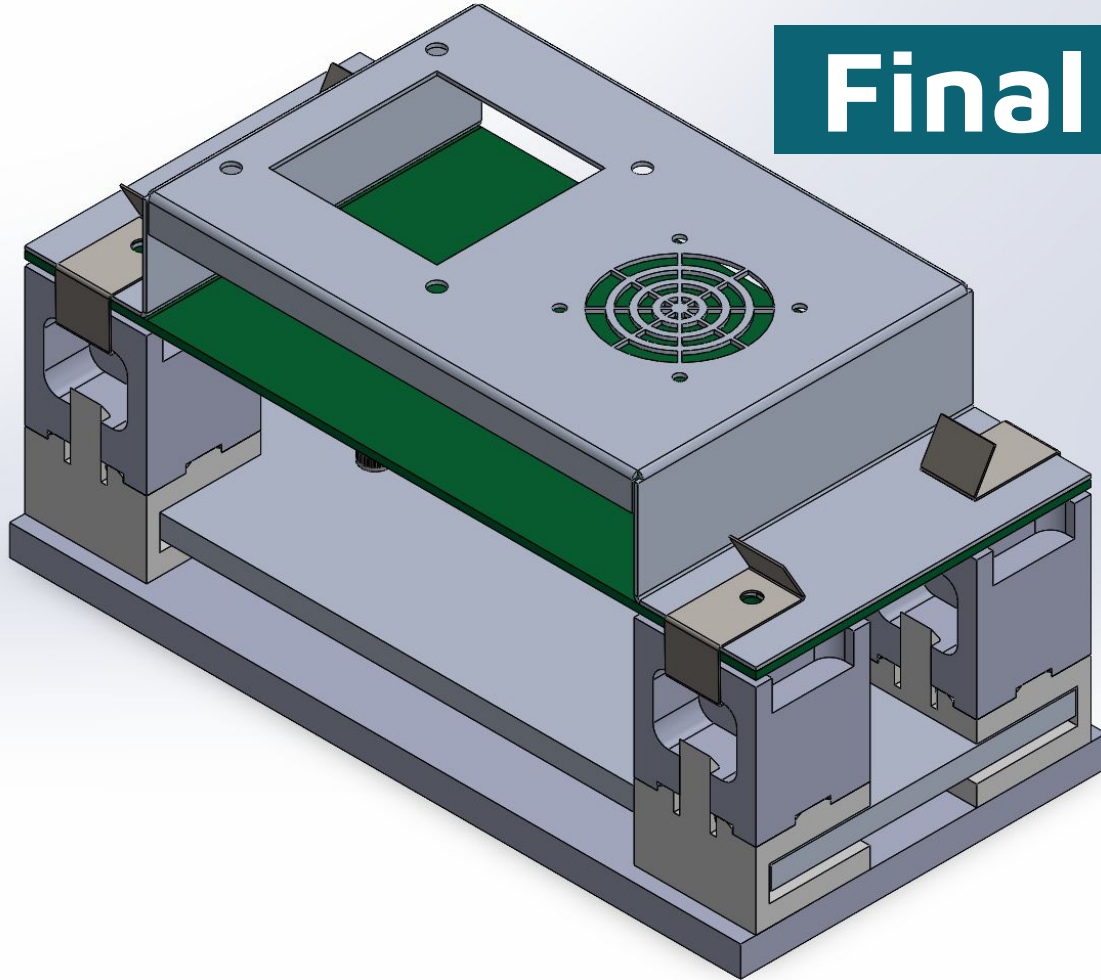


High Level Goals Checklist

- Construct and Assemble a PCB mount assembly that will fasten to an existing baseplate in a piece of equipment
- The mounting parts must hold and locate a PCB and display mounting plate installed without the use of tools

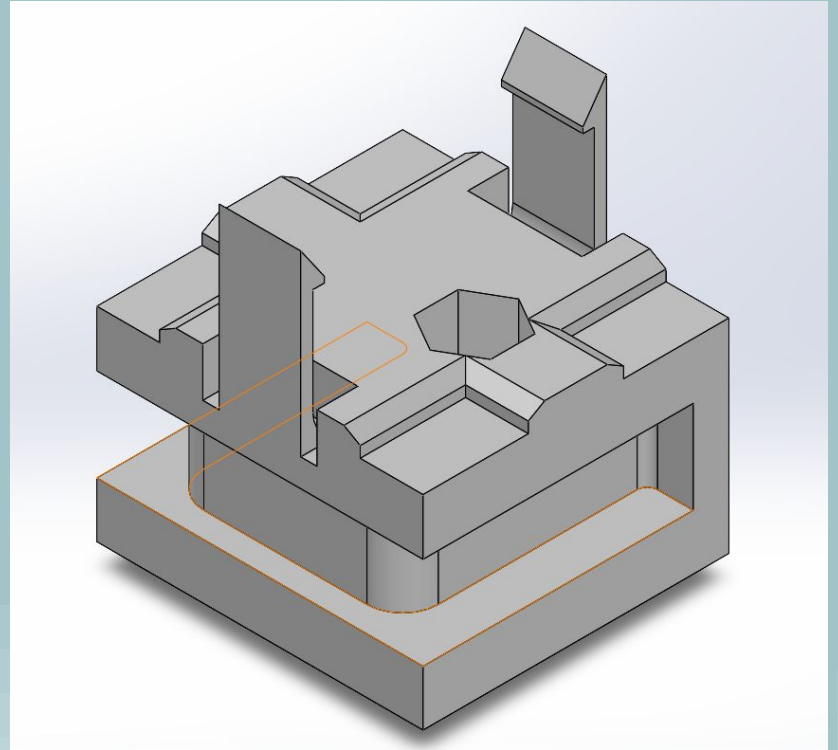


Final Assembly



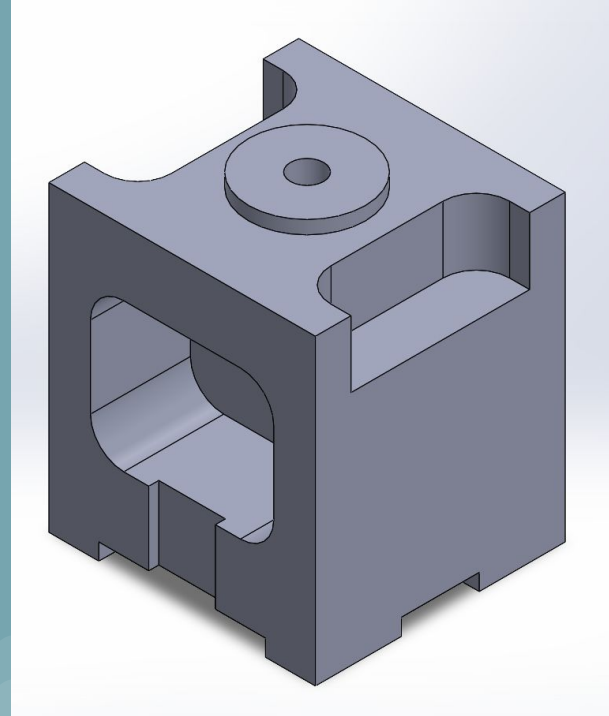
BU01 Clearance Mount

- Meets all functional requirements
- Machine shop printers have best tolerancing
- Cantilever clips have high strength
- Printing material is cheap



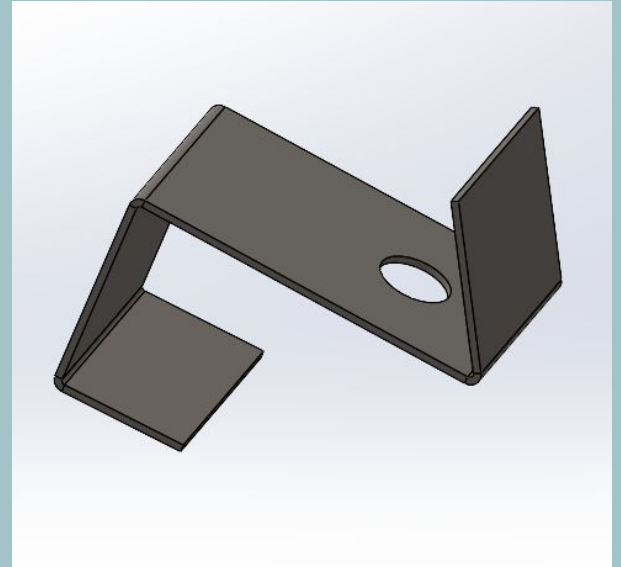
BU02 & BU03 Standoff Mount

- Meets all functional requirements
- Interface with CLM is sturdy and reliable
- During manual assembly, we realized the hole for the dowel pins is slightly shallow



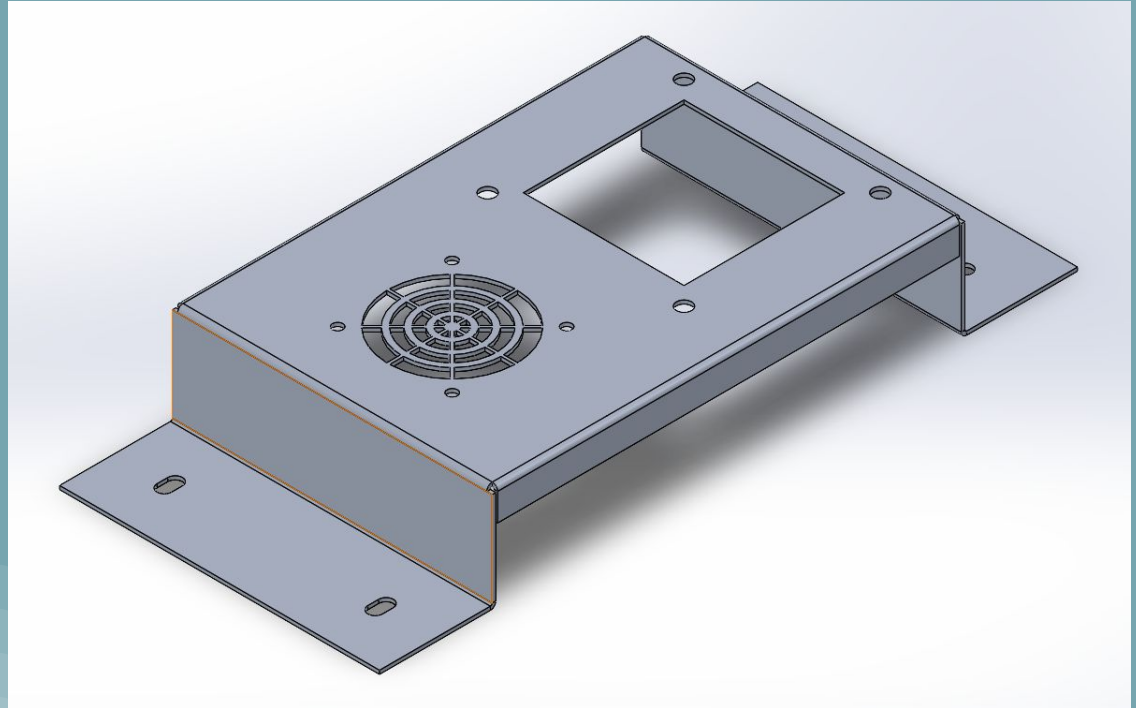
BU04 Sheet Metal Clips

- Prevent PCB board and display plate from moving in z direction
- Very secure
- Disassembly is harder than assembly due to the height of the dowel pin



BU04 Display Mounting Plate

- Fits seamlessly with final assembly
- Slots on one side DFMA
- Slightly longer than given nominal dimension



BU05 & BU06 Fixture & Robotic Assembly



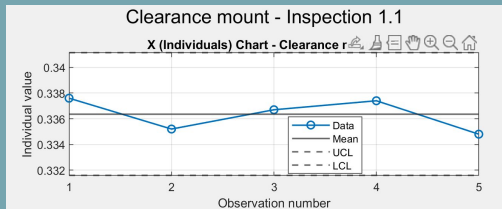
Enterprise Functionality

- Overall process went smoothly!
- Great communication within and between BUs
- Deadlines were met
- Shared Google Drive for organization
- Division of labor was not always equal

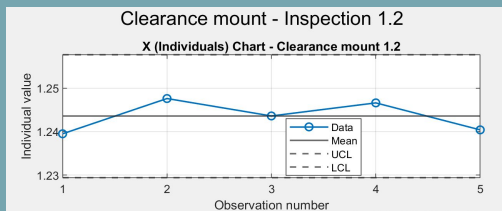
Can We Produce 5000/year Assemblies Consistently?

- We can definitely produce the quantity of units required
- Long but manageable production times for CLM and STM
- We have designed all of our parts for manufacturability

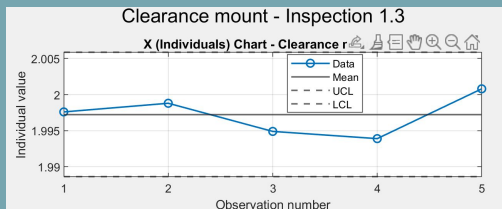
Proven Consistency...



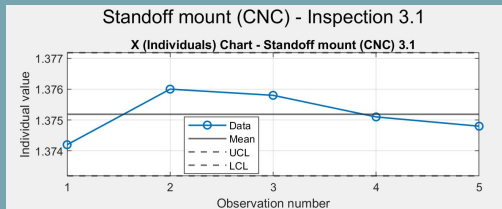
Nominal: 0.3438 Mean: 0.33634 Mean Dev: -0.0074



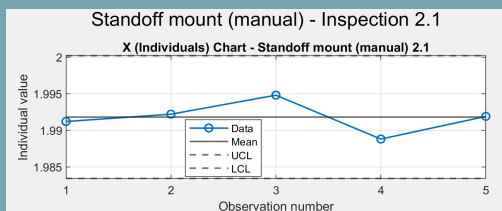
Nominal: 1.2500 Mean: 1.24354 Mean Dev: -0.00646



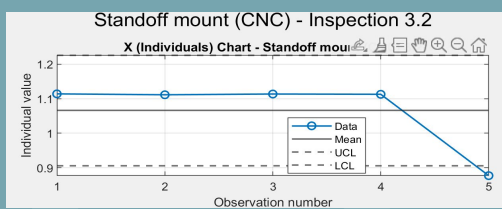
Nominal: 2.0000 Mean: 1.9972 Mean Dev: -0.0028



Nominal: 1.3750 Mean: 1.37518 Mean Dev: 0.00018

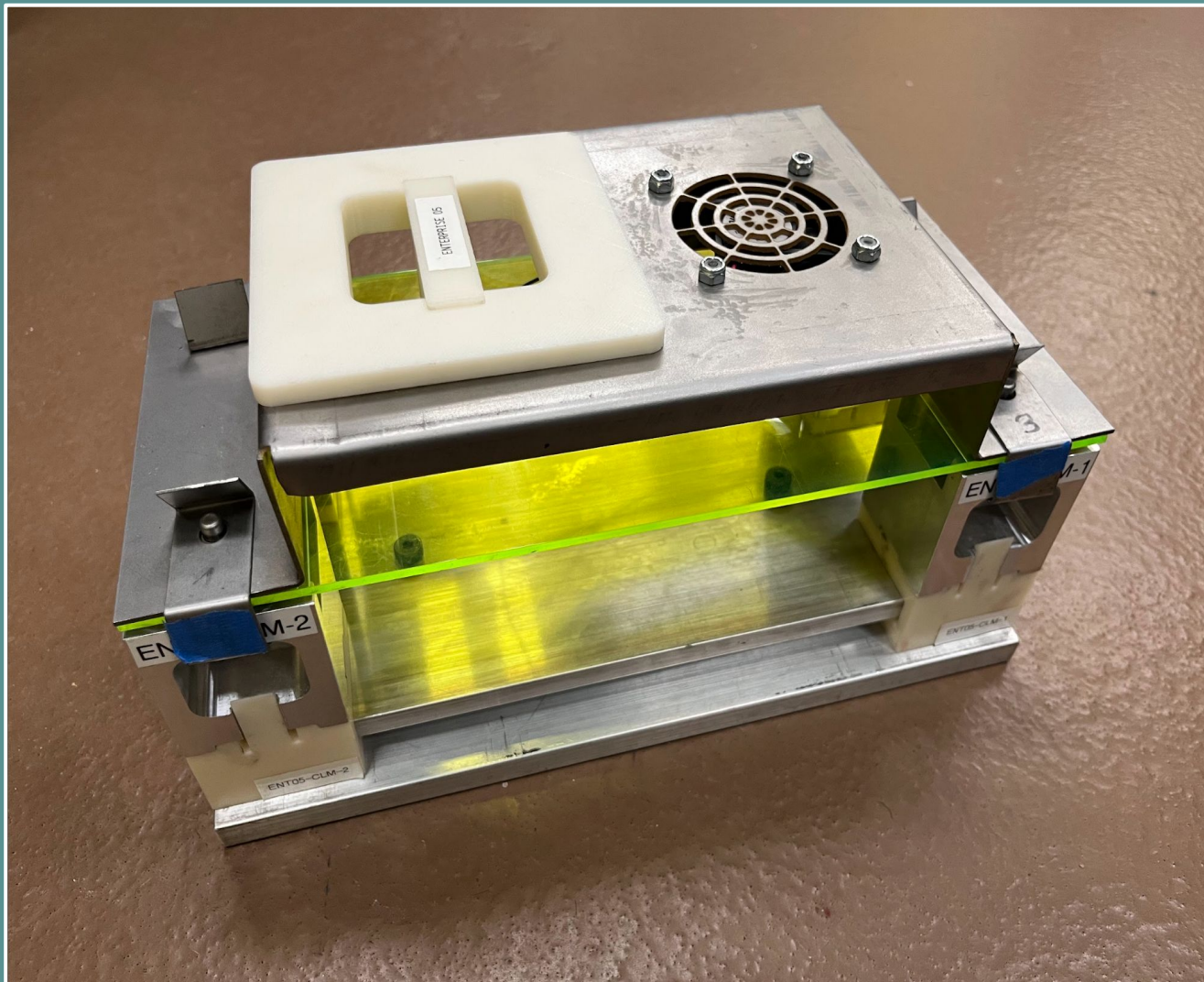


Nominal: 2.0000 Mean: 1.99178 Mean Dev: -0.00822



Nominal: 1.1125 Mean: 1.06578 Mean Dev: -0.04672
Without the outlier... Mean: 1.11295 Mean Dev: 0.00045

- Results well within CLs
- Demonstrates Reliability
- Small changes to recenter data could drive significant results



**Thank you
Enterprise 5!**

Questions

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